

CHIRANJEEVI GUNDU

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SUMMARY

Applied AI Engineer with an M.S. in Computer Science (AI track) and 4 years of professional engineering, with AI running through most of it. I build production Retrieval-Augmented Generation (RAG) systems end to end — document ingestion and structure-aware chunking, embeddings and retrieval tuning, grounded answers with source citations, and LLM evaluation against fixed eval sets — deployed on AWS, Azure, and client-owned on-prem infrastructure. I treat retrieval quality as a measured discipline rather than intuition: separating retrieval failures from generation failures raised retrieval accuracy roughly 35% on a client evaluation set. Backend rigor throughout — typed FastAPI services, PostgreSQL data modeling, automated tests, and CI/CD.

TECHNICAL SKILLS

AI/LLM: Retrieval-Augmented Generation (RAG), Chunking & Retrieval Tuning, Embeddings, Semantic Search, Vector Databases (Pinecone, pgvector), LLM Evaluation & Eval Sets, Grounding & Citation, Prompt Engineering, Context Optimization, Tool-Calling, Multi-Agent Orchestration, OpenAI API, Anthropic Claude, LangChain, LangGraph, LlamaIndex, Hugging Face, LiteLLM, Ollama, Llama 3, Mistral, OpenAI Vision, Guardrails AI, Fine-Tuning Fundamentals (LoRA, QLoRA), Agentic IDEs (Claude Code, GitHub Copilot, Cursor)

Backend & Data: FastAPI, Pydantic, REST APIs, Node.js, React, React Native, Microservices, Service-Oriented Architecture, PostgreSQL 16, pgvector, Pinecone, Redis, Row-Level Security, JWT Auth, Fernet Encryption, SQL Profiling (EXPLAIN/ANALYZE, pg_stat_statements), Indexing & Query Optimization, ETL & Ingestion Pipelines, IMAP, Microsoft Graph

Cloud & DevOps: AWS (EC2, S3, Lambda, ECS), Azure, Docker, Docker Compose, GitHub Actions, CI/CD, Git, PyTest, Plain-SQL Migrations (yoyo), Monitoring & Logging, Audit Trails, Tailscale, Agile/Scrum, Technical Documentation

Languages & ML: Python, Java, SQL, JavaScript, TypeScript, Bash, PowerShell, PyTorch, Scikit-learn, pandas, NumPy, Text Classification

PROFESSIONAL EXPERIENCE

AI Engineer

Aug 2025 – Present

Axitem Software Solution Inc. Remote, USA

- Raised retrieval accuracy roughly 35% by scoring a fixed evaluation set before and after each change and separating retrieval failures from generation failures — the dominant fix was structure-aware parsing that stopped chunking from splitting tables mid-row.
- Deliver production RAG document-Q&A assistants (Python, LangChain, OpenAI APIs) embedded directly inside the portals client teams already use, turning hours of manual document lookup into instant, cited answers now relied on daily by client support and operations teams.
- Ship RAG pipelines (Pinecone vector search, PostgreSQL) that ground every chatbot answer in real company documents with source citations, cutting hallucinations, improving reliability, and earning formal client project sign-off across engagements.
- Automated multi-step business tasks such as data lookup and report drafting with LangChain and LangGraph agent workflows that plan steps, call tools, and verify their own output, removing hours of repetitive manual work for client teams.
- Unlocked privacy-sensitive enterprise deals by running open-source models (Llama 3, Mistral) fully in-house on client-owned infrastructure and clearing security review early — deploying inside the client's own cloud account and walking their IT team through every data flow before writing significant code.
- Cut hallucinations with a three-layer trust design — answers grounded strictly in retrieved chunks, every claim linked to its source passage, and an explicit refusal path when retrieval confidence is low — so the assistant says "not in the documents" instead of guessing.
- Standardized delivery on a Dockerized FastAPI backend template with health endpoints and env-var configuration, making a move between AWS and Azure a configuration change rather than a rewrite and letting new client environments stand up in days.
- Turned every reported wrong answer into a permanent regression case by building evaluation sets from real user questions, then tuning chunking, embeddings, and retrieval against measured hit-rate rather than intuition.
- Keep indexes fresh with scheduled incremental re-ingestion that detects changed documents and upserts only those, versioning chunks so superseded content is removed rather than leaving two conflicting copies of the same policy in retrieval.

Student Worker — Administrative & Marketing Support

Aug 2024 – May 2025

Sodexo Saint Louis, MO (on-campus role during Master's)

- Used AI assistants (ChatGPT, Gemini, spreadsheet AI) daily to compile student dining feedback from online portals, suggestion boxes, and app reviews into themed summaries for management, and to draft social media content for campus dining channels — turning scattered raw comments into recurring themes managers could act on.
- Built spreadsheet formulas and macros to speed daily entry of sales, meal-plan transaction, and temperature-log data into internal tracking systems, and to support inventory and stocking planning, replacing repetitive manual keying.
- Programmed daily digital menu boards across dining halls and retail locations, verifying dietary and allergen labeling (vegan, vegetarian, gluten-free) was accurate before every service.

Backend Software Engineer

Aug 2021 – Jul 2023

Tata Consultancy Services (TCS) Chennai, India

- Cut manual ticket-sorting time 40% for a Fortune 500 client by integrating an ML text-classification model into their IT support system through Python (FastAPI) REST APIs, replacing slow manual triage across busy, high-volume support queues.
- Made the ML routing system safe to switch on by designing a confidence-threshold fallback — low-confidence predictions returned to the manual queue rather than mis-routing — and logging every prediction against the human decision to measure live accuracy.
- Improved mission-critical client workflow performance 60% by profiling and optimizing PostgreSQL queries, adding targeted indexes, and rewriting the inefficient backend data pipelines behind the client's most heavily used reporting workflows.
- Cut deployment failures 30% and held code coverage above 85% by building out thorough PyTest unit-test suites and enforcing CI/CD best practices on every single release shipped to enterprise clients across the delivery account each sprint.
- Diagnosed performance bottlenecks with EXPLAIN/ANALYZE and pg_stat_statements rather than guesswork, then rewrote N+1 access patterns into joins and batch fetches and validated against production-scale data volumes before rollout.
- Translated business-language requirements from Fortune 500 retail and banking stakeholders into scoped technical proposals a non-engineer could approve, establishing the client-facing habit that now anchors my AI delivery work.

PROJECTS

Aadyon Assist — Open-Source AI Life-Ops Platform (github.com/chiranjeevigundu/aadyon-assist)

- Architected and open-sourced (MIT) a self-hosted AI platform end to end — data model, agent layer, ingestion, REST API, and iPhone app — running as seven Docker Compose services on FastAPI and Postgres 16 with pgvector.
- Designed a multi-agent organization in which a CEO agent decomposes a goal and delegates to four specialist agent teams, bounding every run with explicit step limits and recording each step in an audit trail for full traceability and debugging.
- Engineered a human-in-the-loop safety model where read-only analysis runs autonomously while any action with a real-world side effect (money, email) becomes a proposal the user must approve, so no irreversible action ever executes on its own.
- Built a tiered model-routing layer over LiteLLM (OpenRouter, GPT-4o-mini, and local Llama 3 via Ollama) driven entirely by a database table, so cost, capability, and privacy each map to the right model with zero application code changes.
- Designed multi-tenant isolation using Postgres Row-Level Security enforced at the database, with JWT authentication and Fernet-encrypted credentials, so a single bug or injection can never leak one user's data to another user in the system.
- Built ingestion pipelines that turn emails, PDFs, and bank transactions into reviewable to-dos via LLM extraction and OpenAI Vision, routing every extracted item through an approval queue so extraction errors annoy the user instead of corrupting real records.
- Built read-only sync connectors over IMAP and Microsoft Graph with per-account cursors for incremental sync, routing all extracted items through an approval queue so extraction errors annoy the user instead of corrupting real records.
- Made a solo project survivable months later by concentrating 149 PyTest cases where determinism lives — schema and CRUD, auth and Row-Level Security boundaries, scoring math, API contracts — and testing around LLM calls rather than asserting on generations, with GitHub Actions CI and plain-SQL migrations.
- Deployed the whole stack self-hosted and reachable only over a Tailscale network, with a daily briefing pushed to iPhone via self-hosted ntfy, making privacy structural rather than a policy promise.

EDUCATION

Saint Louis University, St. Louis, MO — M.S. Computer Science (Artificial Intelligence Track)

Aug 2023 – May 2025

Jawaharlal Nehru Technological University — B.Tech, Electronics & Communication Engineering

Jul 2017 – Aug 2021